Integrador

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$$SNR_2 = SNR_1 - 20 \cdot \log(K)$$

Luego si $SNR_2 = 0 \Rightarrow$ $10 \cdot \log\left[\frac{P_S}{P_R}\right] = 20 \cdot \log(K)$

$$K^2 = \frac{P_S}{P_R} \Rightarrow K = \sqrt{\frac{P_S}{P_R}}$$

Si $SNR_2 = 6 \Rightarrow$ $6 = 10 \cdot \log\left[\frac{P_S}{P_R}\right] - 20 \cdot \log(K)$

$$-6 = -10 \cdot \log\left[\frac{P_S}{P_R}\right] + 20 \cdot \log(K)$$

$$-6 + 10 \cdot \log\left[\frac{P_S}{P_R}\right] = 20 \cdot \log(K)$$

$$\frac{-6}{10} + \log\left[\frac{P_S}{P_R}\right] = \log(K^2)$$

$$10^{-\frac{6}{10} + \log\left[\frac{P_S}{P_R}\right]} = K^2$$

$$10^{-\frac{6}{10}} \frac{P_S}{P_R} = K^2$$

$$10^{-\frac{6}{20}} \cdot \sqrt{\frac{P_S}{\ }} = K$$

$$10^{-\frac{b}{20}} \cdot \sqrt{\frac{P_S}{P_R}} = K$$